Statistical Machine Learning: DC Urban Forestry

Dataset Background, EDA, and Modeling Approaches

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Dataset Background: Data Source

- **Dataset:** DC Urban Forestry Street Trees (`Urban_Forestry_Street_Trees.csv`)
- **Source:** Open Data DC / DC Urban Forestry Division
- **Unit of observation:** one street tree (identified by `OBJECTID` / `GLOBALID`)
- **Size:** 222830 x 47

Dataset Variables

- **Contents:**
- Tree attributes: species (`SCI_NM`, `GENUS_NAME`), size (`DBH`), `CONDITION`
- Site attributes: tree box size (`TBOX_L`, `TBOX_W`), `WARD`, `CURB`, `SIDEWALK`, `WIRES`
- Lidar-derived crown measurements (2020, in feet): `MBG_WIDTH`, `MBG_LENGTH`, `MAX_MEAN`, `MAX_CROWN_HEIGHT`
- Administrative fields: `CREATED_DATE`, `LAST_EDITED_DATE`, `RETIREDDT`
- Location: `X`, `Y` coordinates

Dataset Background: Variables of Interest

- **Possible response:** `CONDITION` (categorical)
- Collapse into “good” vs. “bad” for logistic regression
- Estimate the probability a tree is in good condition
- **Numeric predictors:** `DBH`, `TBOX_L`, `TBOX_W`, `MBG_WIDTH`, `MAX_CROWN_HEIGHT`
- **Categorical predictors:** `GENUS_NAME`, `WARD`, `WIRES`, `CURB`, `SIDEWALK`

Exploratory Data Analysis: Condition Obstacles

- **Retired vs. living trees** (`RETIREDDT`)
- Splits the data into two populations
- Only 5,452 trees have a retirement date

Missing/Invalid Data

- Several columns have > 100,000 NAs *(list them from the NA table)*
- Missingness for the model variables checked as % missing (CONDITION, predictors)
- **Invalid values**
- Negative DBH values present
- Extreme DBH values > 100 inches
- Extreme TBOX_L outlier (one record flagged for review)
- **Cleaning decision:** DBH restricted to 0 to 100 for plots, **decide about modeling**.

Exploratory Data Analysis (pt 1): Numeric Variables

Exploratory Data Analysis (pt 2): Numeric Variables

Exploratory Data Analysis (pt 1): Relationships and Spatial View

- **Correlation among numeric predictors**
- `MBG_WIDTH` and `MBG_LENGTH` correlation: [value from `cor(forestry.pairs)`]
- Collinearity check on `DBH`, `TBOX_L`, `TBOX_W`, `MBG_WIDTH`, `MAX_CROWN_HEIGHT`
- `MBG_WIDTH` is the main concern; no other pair above ~0.6
- **Spatial distribution:** `X` vs. `Y` scatter shows tree locations across the city

Machine Learning Algorithm #1

- **Overview:** *algorithm name + what it does*
- **Assumptions:**
- **Why it fits this data:**
- **Setup / results:**

Machine Learning Algorithm #2

- **Overview:** *algorithm name + what it does*
- **Assumptions:**
- **Why it fits this data:**
- **Setup / results:**

Machine Learning Algorithm #3

- **Overview:** *algorithm name + what it does*
- **Assumptions:**
- **Why it fits this data:**
- **Setup / results:**

Machine Learning Algorithm #4

- **Overview:** *algorithm name + what it does*
- **Assumptions:**
- **Why it fits this data:**
- **Setup / results:**

Speaker notes